

An Internship Report
Of
CodSoft Frontend Development Virtual Internship

Submitted
To
**School of Computer Science and Engineering IILM University, Greater Noida,
U.P.**

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By
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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, ABHINAV KUMAR, do hereby declare that the internship report titled “CodSoft Frontend Development Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ABHINAV KUMAR

Roll No.: 2410031432

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank CodSoft for providing me the opportunity and the structured, project-based platform to learn and gain hands-on exposure in the domain of Frontend Development. I am also thankful to my institute, IILM University, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Student Name: ABHINAV KUMAR

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TABLE OF CONTENTS

1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Completion Certificate	-
4. Project Description	1
4.1 Introduction	1
4.2 Organization Profile	1
4.3 Problem Statement	2
4.4 Project Objectives.....	2
4.5 Scope of the Project.....	3
4.6 Technologies and Tools Used	3
4.7 System Architecture	4
4.8 Methodology	4
4.9 Expected Outcomes.....	5
4.10 Certificates of Completion and Communication Proof.....	6
5. Bibliography/References	7

3. INTERNSHIP COMPLETION CERTIFICATE



Certificate of Completion — CodSoft Frontend Development Internship (C.ID: e431cb6)

INTERNSHIP OFFER LETTER

Date : 14/07/2026

ID:BY26RY220140

Dear,

Abhinav Kumar

We would like to congratulate you on being selected for the "Frontend Development" virtual internship position with "CodSoft". We at CodSoft are excited that you will join our team.

The duration of the internship will be of **1 Month**, starting from **15 July 2026 to 15 August 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects.

As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Thank You!



Founder



MSME Registered



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 4-week Virtual Internship undertaken under the CodSoft Virtual Internship Program, in the domain of “Frontend Development”. The internship was carried out from 15 July 2026 to 15 August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University.

The primary aim of this internship was to build a strong practical foundation in client-side web development using HTML5, CSS3 and JavaScript, and to apply it to real-world application problems such as responsive portfolio design, task management, personal finance tracking, media playback and content publishing.

The internship followed a structured, task-based curriculum combining conceptual learning with hands-on project work, version-controlled submissions and video-based demonstrations. Interns were required to complete a minimum of three tasks out of five specified tasks for successful completion. On successful completion of all the requirements, a Certificate of Completion (C.ID: e431cb6) was awarded by CodSoft, with remarks noting the candidate's showcased skills and contributions to the tasks and projects undertaken during the internship.

4.2 Organization Profile

CodSoft is an ISO 9001:2015 certified and MSME-registered technology education and software development platform designed to empower aspiring developers. Its stated goal is to transform theoretical knowledge into practical expertise.

The organization operates through virtual internships, AI-driven career tools and project-based learning curriculums, together forming an ecosystem for student growth. Its career-tool suite includes a Resume Builder, ATS Score Checker, AI Interview Coach, Career Roadmap Generator, Portfolio Builder, one-to-one mentoring sessions and interview preparation modules. CodSoftAI, the organization's in-house assistant, is positioned as a round-the-clock mentor, resume reviewer and technical guide. Interns additionally receive a completion certificate, networking opportunities and placement support.

4.3 Problem Statement

Although a large number of engineering students learn the syntax of HTML, CSS and JavaScript during their academic coursework, very few get structured, hands-on exposure to building complete, deployable web applications. Concepts such as responsive layout, client-side state management, data persistence and form validation are often understood only at a theoretical level, with little practice in assembling them into working products that behave correctly across devices and browsers.

The problem addressed by this internship was to practically learn and apply the complete frontend development workflow — from interpreting a written requirement specification, to designing a responsive interface, to implementing interactive behaviour and persistent client-side data, and finally to version-controlling and publishing the finished application.

4.4 Project Objectives

- To understand and apply the fundamentals of frontend development by building fully responsive web interfaces using HTML5, CSS3 and JavaScript.
- To design and develop a modern personal portfolio website presenting skills, projects and professional profile.
- To implement responsive navigation, smooth scrolling, hover animations and other interactive user-interface elements.
- To perform client-side form validation before data submission.
- To build a task management application supporting complete create, read, update and delete operations.
- To implement client-side data persistence using the Web Storage API so that application data survives a page refresh.
- To develop an expense tracking application with automatic computation of income, expenditure and balance.
- To implement search, filtering and category-based organization of records.
- To develop a web-based music player using the HTML5 audio element and its JavaScript media API.
- To build a responsive blog website with article cards, detail pages, search and pagination.

- To maintain structured GitHub repositories and demonstrate the completed work through recorded project videos.

4.5 Scope of the Project

The scope of this internship spans the complete client-side web development workflow — starting from understanding markup and layout fundamentals and building responsive interfaces, moving to interactive behaviour through DOM manipulation and event handling, and extending into client-side data persistence using Local Storage. It further covers media handling through the HTML5 audio API, content-driven page generation for a blog application, and version control and static deployment through GitHub and GitHub Pages.

The internship was limited to frontend, browser-based development. It did not involve server-side programming, database administration or deployment on any live organizational production system. All application data was stored on the client device, and no backend services were integrated.

4.6 Technologies and Tools Used

- **Markup and Structure:** HTML5, semantic elements, accessible form markup.
- **Styling and Layout:** CSS3, Flexbox, CSS Grid, media queries, transitions, keyframe animations and CSS custom properties for theming.
- **Scripting:** JavaScript (ES6+) — DOM API, event listeners, event delegation, array methods, template literals and arrow functions.
- **Client-side Persistence:** Web Storage API (localStorage) with JSON serialisation.
- **Media:** HTML5 audio element and its associated JavaScript media API.
- **Formatting and Internationalisation:** Intl.NumberFormat for currency display.
- **Version Control:** Git and GitHub, with repositories maintained under the CODSOFT naming convention.
- **Editor and Testing Tools:** Visual Studio Code, Live Server, Chrome DevTools (device toolbar, console, Lighthouse audits).
- **Deployment:** GitHub Pages static hosting.

4.7 System Architecture

The general architecture followed during the internship for the frontend development track can be summarized in the following layers:

- **Presentation Layer:** Semantic HTML5 markup defines the structure of each application, styled through CSS3 using Flexbox and Grid, with media queries providing responsive adaptation across device widths.
- **Interaction Layer:** JavaScript event listeners capture user actions such as clicks, keystrokes, form submissions and media events, and dispatch them to the appropriate handlers.
- **Logic Layer:** Application state is held in JavaScript arrays and objects acting as a single source of truth; all derived values, such as totals and counters, are recomputed from this state rather than stored separately.
- **Validation Layer:** User input is validated and sanitised before it is admitted into application state, with inline error messaging returned to the interface.
- **Persistence Layer:** Application state is serialised to JSON and written to the browser's Local Storage after every mutation, and parsed back on page load with fallback handling for absent or corrupt data.
- **Rendering Layer:** A single render routine rebuilds the affected portion of the DOM from current state, ensuring that the interface and the data can never fall out of step.
- **Deployment Layer:** Completed builds are committed to task-specific GitHub repositories and published as static sites through GitHub Pages.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on project development and self-evaluation against the published requirement specification, as summarized in the table below.

Week	Module / Task	Description
1	Task 1 — Personal Portfolio Website	Foundations of responsive design; building a multi-section portfolio website with Home, About, Skills, Projects, Resume and Contact sections, project cards, smooth scrolling, hover animations, a mobile navigation menu, a downloadable CV button and a validated contact form.
2	Task 2 — To-Do List Application	Building a responsive task manager supporting addition with validation, editing, deletion and completion toggling of tasks; persistence through Local Storage; live completed and pending

Week	Module / Task	Description
		counters; search and filtering. Bonus features included categories, priority levels, due dates and dark mode.
3	Task 3 — Expense Tracker	Developing a personal finance application allowing income and expense entry, transaction history with date and category, automatically derived totals and balance, editing and deletion of records, category-based filtering and summary cards, with all data stored locally.
4	Tasks 4 and 5 — Music Player and Blog Website	Developing a web-based music player with play, pause, next and previous controls, a seekable progress bar, volume and mute control and live playback time; and a responsive blog website with featured posts, article cards, category and tag organization, detail pages, search and a Load More control.
Final	Submission and Evaluation	Publishing each project to a dedicated GitHub repository, recording and posting project demonstration videos, submitting repository links through the official task submission form, and issue of the Certificate of Completion.

Note: There was no stipend associated with this internship.

Each task concluded with a self-assessment against the published requirement checklist, cross-browser and multi-device testing using browser developer tools, and a recorded demonstration of the working application. The minimum requirement for successful completion of the internship was the satisfactory delivery of three tasks.

4.9 Expected Outcomes

- Ability to design and build fully responsive web interfaces using HTML5, CSS3, Flexbox, CSS Grid and media queries, adapting correctly from mobile to desktop viewports.
- Practical understanding of DOM manipulation and event-driven programming, including event delegation for dynamically generated elements.
- Capability to implement complete create, read, update and delete functionality entirely on the client side.
- Working knowledge of the Web Storage API, including its practical limits — synchronous access, a storage ceiling of a few megabytes, string-only values, and confinement to a single browser on a single device.
- Skill in validating and sanitising user input before it enters application state, with helpful inline error reporting.

- Hands-on exposure to the HTML5 media API for building a custom audio player interface.
- Competence in version control using Git and GitHub, and in publishing working builds through static hosting.
- A portfolio of deployable web applications that can be demonstrated to prospective employers, together with a Certificate of Completion issued by CodSoft.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by CodSoft, confirming successful completion of the 4-week Virtual Internship Program in Frontend Development, is attached in Chapter 3 of this report. The certificate carries Certificate ID e431cb6 and was issued on 06 August 2026 under the signature of the Founder, CodSoft.

The GitHub repositories containing the source code of the completed tasks, screenshots of the deployed applications, and the project demonstration videos posted on the professional networking platform are submitted as further communication proof. Repository links and video links are to be inserted below.

GitHub Repository Links: https://github.com/yabhinav997/CODSOFT_TASKSNO1-TASKNO5.git

5. BIBLIOGRAPHY/REFERENCES

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9. GitHub Pages Documentation, <https://docs.github.com/en/pages>
10. Official CodSoft Website, <https://www.codsoft.in>